

PRESIDENTE PRUDENTE, BrazilWMO: **837160**Lat: **22.175S**Lon: **51.425W**Elev: **450**StdP: **96.03**Time Zone: **-3.00 (SAB)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	9.1	11.1	1.0	4.3	20.5	3.2	5.0	21.6	9.1	22.0	8.3	19.9	3.0	230	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.7	36.0	20.7	34.9	21.0	33.9	21.3	25.2	30.4	24.7	29.8	24.3	29.2	3.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	20.0	26.7	23.2	19.0	26.1	23.1	18.8	26.0	79.5	30.3	77.5	30.2	75.9	29.1	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.7	7.0	4.8	37.7	2.2	1.4	3.2	38.7	1.9	39.5	0.6	40.2	-1.0	41.2		
(5)			4.0	25.7	2.2	0.9	2.5	26.3	1.2	26.9	0.0	27.4	-1.6	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.2	26.6	26.6	26.0	24.4	21.6	20.6	20.8	22.6	24.2	25.2	25.5	26.5
	DBStd	3.50	1.86	1.78	1.78	2.43	2.91	3.18	3.67	3.60	3.66	2.80	2.14	1.82
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD18.3	59	0	0	0	1	8	16	22	9	3	0	0	0
	CDD10.0	5180	513	466	496	431	359	317	336	389	426	472	464	511
	CDD18.3	2196	255	233	237	182	108	83	99	140	179	213	214	252
	CDH23.3	19158	2183	2002	2034	1435	711	542	754	1289	1821	2106	1987	2293
CDH26.7	8001	927	844	836	541	207	139	230	548	876	994	856	1003	
Wind	WSAvg	3.4	3.2	2.9	3.0	3.2	3.2	3.4	3.6	4.0	4.1	3.8	3.5	3.2
Precipitation	PrecAvg	1385	248	188	135	79	79	50	36	41	86	121	133	176
	PrecMax	2065	532	354	294	173	204	254	247	151	253	249	277	345
	PrecMin	1127	70	62	42	4	14	0	3	0	0	29	56	77
	PrecStd	207	121	69	63	44	53	51	50	45	67	61	51	73
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	36.0	34.9	33.2	32.0	31.1	32.0	35.0	37.1	37.2	35.2	36.0
		MCWB	22.0	21.3	22.6	22.2	20.9	18.4	18.5	17.4	18.7	20.4	22.0	22.0
	2%	DB	34.2	34.2	33.4	32.2	30.6	29.8	30.8	33.1	35.2	35.9	34.1	34.2
		MCWB	22.5	22.6	22.7	21.9	20.2	18.8	18.0	17.9	18.8	20.8	21.4	22.1
	5%	DB	33.0	33.0	32.2	31.1	29.1	28.2	29.1	31.8	33.9	34.0	32.9	33.1
		MCWB	22.9	22.9	22.8	21.3	19.7	18.6	17.8	17.8	18.9	21.1	21.5	22.1
	10%	DB	31.6	31.6	31.0	30.0	27.3	26.9	27.9	30.0	31.9	32.0	31.2	31.9
		MCWB	22.9	23.2	22.7	21.3	19.3	18.4	17.6	17.7	19.1	21.0	21.5	22.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	25.7	25.3	24.8	22.5	22.4	21.0	21.9	23.3	24.2	25.6	25.7
		MCDB	30.5	31.3	30.5	30.4	27.9	26.7	26.9	28.3	31.0	30.7	31.5	30.0
	2%	WB	24.8	25.0	24.6	23.9	21.7	21.2	20.0	20.4	22.3	23.3	24.5	25.0
		MCDB	30.0	30.4	29.9	29.1	27.4	25.6	26.5	26.2	28.7	29.8	29.8	29.6
	5%	WB	24.2	24.5	24.1	23.2	21.0	20.1	19.3	19.5	21.4	22.7	23.7	24.3
		MCDB	29.1	29.5	29.4	28.1	26.2	24.6	25.8	26.3	27.7	29.1	28.7	28.8
	10%	WB	23.8	24.0	23.5	22.6	20.2	19.3	18.6	18.6	20.6	22.0	23.0	23.7
		MCDB	28.3	28.6	28.6	27.1	24.8	23.9	24.9	26.2	26.8	28.3	27.8	28.1
Mean Daily Temperature Range	5% DB	MDBR	9.7	9.7	10.1	10.7	10.6	10.7	11.9	12.6	12.6	11.7	11.2	10.4
		MCDBR	11.8	11.7	11.6	11.8	12.0	12.0	12.9	14.3	14.5	14.4	13.2	12.5
		MCWBR	3.4	3.5	3.7	4.1	4.4	4.4	4.7	5.0	4.5	4.4	4.4	3.9
	5% WB	MCDBR	9.5	9.8	9.9	10.0	10.3	10.2	11.4	12.1	11.8	11.2	10.2	10.0
		MCWBR	3.6	3.6	3.8	4.1	4.5	4.8	5.1	5.1	4.9	4.4	4.3	3.9
Clear-Sky Solar Irradiance	taub		0.393	0.396	0.388	0.377	0.358	0.368	0.368	0.422	0.538	0.467	0.413	0.401
	taud		2.449	2.431	2.448	2.461	2.469	2.443	2.421	2.225	1.913	2.181	2.338	2.399
	Ebn at Noon		952	935	917	884	862	829	842	826	769	861	926	945
	Edh at Noon		122	122	115	105	96	95	99	131	193	155	135	128
All-Sky Solar Radiation	RadAvg		5.90	5.97	5.61	5.00	3.98	3.68	3.93	4.73	5.14	5.66	6.19	6.24
	RadStd		0.59	0.39	0.40	0.31	0.35	0.29	0.35	0.35	0.49	0.39	0.41	0.51